

Quantization of the domain-wall memory effect on a spin chain, without scattering assumptions

A self-contained report

Project `tns-as-st-me` — campaign report*

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Abstract

A wall between two oppositely magnetized halves of a spin chain shifts its position when a spin wave scatters off it, and the shift survives after the spin wave has left: the chain remembers the collision. This report proves, in a precisely delimited sense, that this remembered shift is *quantized* — each recorded outcome is an integer multiple of a unit fixed by the magnetization density — and that the proof needs *no* assumptions about how the scattering proceeds: no decomposition into reflected and transmitted waves, no absence of bound states, no completeness of any scattering theory. Those scattering assumptions, previously believed necessary, are needed only to compute *which* multiple occurs with which probability, not to prove that only integer multiples occur. Three further results are established along the way: the average shift itself is *not* quantized (it is a probability-weighted average of quantized outcomes, and the distinction is essential); a by-product theorem shows that the magnetization density of a symmetric matrix-product ground state is itself quantized in half-integers; and there is provably *no* single operator whose spectrum expresses the quantization — the measurement protocol in the statement cannot be removed. Finally, for chains whose excitations are described exactly by the matrix-product excitation ansatz, the existence half of the scattering theory is proved from the ansatz data. Every term used above is defined in the body of the report. Each theorem’s exact hypotheses, including two assumptions that remain unproved from first principles, are stated in full; a section lists what has *not* been proved. All results were verified by adversarial review and by machine-checked certificates, including a pre-registered numerical falsification attempt that failed to falsify.

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*Prepared 2026-08-28 by the project’s automated adversarial-verification campaign. Every theorem below was produced by a proposer–critic loop in which independent provers and hostile reviewers (from different model families) attacked each statement until no fatal or major objection remained; the final adjudication records are cited in Appendix A. The internal claim identifiers (`M-INDEX-fin`, `M-INDEX-spec`, `M-IDX-density`, `AC-EX`) refer to the project’s claim ledger.

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1 Summary of results

The whole content of this report in one sentence: *on an infinite spin chain carrying a wall between two domains of opposite magnetization density $\pm\rho$, the outcome of measuring the wall’s displacement — by the explicit two-measurement protocol of Definition 6.2 — is always an integer multiple of $1/(2\rho)$ lattice sites, provided only that the chain’s local dynamics conserves magnetization and that the measured region settles down at late times (Hypothesis 6.3); no property of the scattering process is assumed.*

The five main statements, with their logical status:

	status	content
Theorem 7.1	proved, unconditional	at any fixed finite window, the measured charge lies on a rigid integer lattice, at all times
Theorem 7.2	proved from Hyp. 2.5+6.3	the asymptotic outcome distribution of the two-measurement protocol is supported on the integers; the mean shift is $\delta x = -(2\rho)^{-1} \sum_{\nu} \nu p_{\nu}$
Theorem 3.5	proved, unconditional	the density itself obeys $2\rho \in \mathbb{Z}$ for symmetric injective matrix-product vacua
Theorem 5.1	proved (a refutation)	no global charge operator exists whose spectrum could express Theorem 7.2; the protocol is unremovable
Theorem 8.3	proved from Hyp. 8.2	scattering states (wave operators) <i>exist</i> whenever the matrix-product excitation ansatz is exact; their completeness is not claimed and not needed

Section 10 lists, without euphemism, what is assumed and what remains open. Section 9 reports the numerical checks, including a falsification attempt that was specified before the theorems were proved.

2 The setting

Everything in this report happens on an infinite one-dimensional lattice of quantum spins. This section defines the lattice, its observables, its states, and its dynamics. A reader familiar with

quantum spin systems can skim to Section 3; nothing nonstandard appears before then.

Definition 2.1 (spin chain, local observables). Fix an integer $d \geq 2$ (the *on-site dimension*). To each site $x \in \mathbb{Z}$ attach the Hilbert space $\mathbb{C}^d$. For a finite set $\Lambda \subset \mathbb{Z}$, an *observable supported in Λ* is a linear operator on $\bigotimes_{x \in \Lambda} \mathbb{C}^d$, extended to act as the identity on all other sites. The collection of all such operators, for all finite Λ , together with the limits of sequences that converge in operator norm, is denoted $\mathcal{A}$ and called the *observable algebra* of the chain.

Definition 2.2 (state). A *state* ω is a linear map $\omega : \mathcal{A} \rightarrow \mathbb{C}$ assigning to each observable its expectation value, with $\omega(A^\dagger A) \geq 0$ for all A and $\omega(\mathbf{1}) = 1$. This is the standard notion: a state is a consistent assignment of expectation values to all local measurements.

Definition 2.3 (Hilbert-space realization). Every state ω can be *realized* on a Hilbert space: there exist a Hilbert space $\mathcal{H}$, a unit vector $\Psi \in \mathcal{H}$, and an assignment π of an operator $\pi(A)$ on $\mathcal{H}$ to each $A \in \mathcal{A}$, compatible with sums, products and adjoints, such that $\omega(A) = \langle \Psi, \pi(A)\Psi \rangle$ for all $A \in \mathcal{A}$. (This is the Gelfand–Naimark–Segal construction; we use only its existence.) When a state is fixed we write A for $\pi(A)$.

Definition 2.4 (on-site charge, total charge of a region). Fix a Hermitian $d \times d$ matrix S^z (the *on-site charge*; for a spin- S chain, $d = 2S + 1$ and S^z is the usual third spin component with eigenvalues $-S, -S + 1, \dots, S$). Write S_x^z for the copy of S^z at site x . For a finite interval $W = [a, b] \subset \mathbb{Z}$ the *charge of W* is $\sum_{x=a}^b S_x^z$. The one-parameter family of unitaries $u(\theta) = e^{i\theta S^z}$, applied simultaneously at every site of a region, implements a *rotation* of that region by angle θ .

Hypothesis 2.5 (integer spacing of the on-site charge, (INT)). There is a complex number c with $|c| = 1$ such that $e^{2\pi i S^z} = c \mathbf{1}$ on one site. Equivalently, writing $c = e^{2\pi i \kappa}$ with $\kappa \in [0, 1)$: all eigenvalues of S^z lie in the shifted integer lattice $\kappa + \mathbb{Z}$.

Remark 2.6. Hypothesis 2.5 holds for every spin- S chain, with $c = e^{2\pi i S}$: the eigenvalues of the third spin component are spaced by exactly 1. It is the only structural input about the on-site charge that any result below uses. No result assumes anything about the *value* of any density; densities are outputs (Theorem 3.5), not inputs.

Definition 2.7 (dynamics). A *finite-range, charge-conserving Hamiltonian* is a formal sum $H = \sum_{x \in \mathbb{Z}} h_x$ where each h_x is a fixed Hermitian observable supported in $[x, x + R]$ for some fixed range $R < \infty$, independent of x up to translation, and each h_x commutes with the total charge of any interval containing its support plus a collar of width R . Such a sum generates, in a standard way, a family of transformations α_t ($t \in \mathbb{R}$) of the observable algebra: $\alpha_t(A)$ is the observable A evolved for time t (Heisenberg picture). Charge conservation means concretely: the change of the charge of W in time equals the net current through the two ends of W , where the *current* at a bond is itself a local observable determined by H ; no charge is created or destroyed in the interior.

3 The two vacua, matrix product states, and the density theorem

The wall of the title separates two translation-invariant reference states. This section defines the class of reference states used — matrix product states — and proves the first main theorem, which constrains their densities. The definitions here are standard in the tensor-network literature [2]; they are restated so that this report is self-contained.

Definition 3.1 (matrix product state, MPS). Fix an integer $\chi \geq 1$ (the *bond dimension*) and a collection $A = (A^1, \dots, A^d)$ of $\chi \times \chi$ complex matrices, one per basis state of the site Hilbert space $\mathbb{C}^d$. Define the linear map $\mathbb{E}$ on $\chi \times \chi$ matrices, and for each single-site operator X the map $\mathbb{E}_X$, by

$$\mathbb{E}(Y) := \sum_{m=1}^d A^m Y (A^m)^\dagger, \quad \mathbb{E}_X(Y) := \sum_{m,n=1}^d \langle n|X|m \rangle A^m Y (A^n)^\dagger.$$

Assume the normalization: the largest eigenvalue of $\mathbb{E}$ (in absolute value) is 1, it is the unique eigenvalue on the unit circle, and it has positive-definite fixed points — a matrix Λ with $\mathbb{E}(\Lambda) = \Lambda$ and a matrix Λ' with $\mathbb{E}^\dagger(\Lambda') = \Lambda'$, normalized by $\text{tr}(\Lambda' \Lambda) = 1$. The *matrix product state* ω_A is the translation-invariant state whose expectation of a product $X_1 \otimes \dots \otimes X_n$ of single-site operators on n consecutive sites is

$$\omega_A(X_1 \otimes \dots \otimes X_n) = \text{tr}(\Lambda' \mathbb{E}_{X_1} \circ \dots \circ \mathbb{E}_{X_n}(\Lambda)).$$

The matrices A^m act on an auxiliary χ -dimensional space (the *bond space*) that carries the correlations of the state; $\chi = 1$ gives exactly the uncorrelated product states.

Definition 3.2 (injectivity). The tensor A is *injective* if there is a length k such that the products $A^{m_1} A^{m_2} \dots A^{m_k}$, over all choices of indices, span the full space of $\chi \times \chi$ matrices. Injectivity is the genericity condition of the MPS literature: it guarantees that the state determines the tensor up to the natural equivalences, and that the normalization of Definition 3.1 can be arranged. All reference states below are injective MPS.

Definition 3.3 (symmetric MPS pair with opposite densities). Fix two injective MPS tensors A_α, A_β of a common bond dimension χ , with states $\omega_\alpha, \omega_\beta$ (the two *vacua*). We assume:

1. (*invariance*) both states are invariant under all rotations: $\omega_\gamma(u(\theta)^{\otimes n} B (u(\theta)^\dagger)^{\otimes n}) = \omega_\gamma(B)$ for $\gamma \in \{\alpha, \beta\}$, every θ , and every observable B on n sites;
2. (*local implementation*) for each vacuum there are invertible $\chi \times \chi$ matrices V_θ and phases $f(\theta)$, with $V_0 = \mathbf{1}$, $f(0) = 0$, both differentiable in θ , such that rotating the physical index of the tensor equals a similarity transformation on the bond space:

$$\sum_n [u(\theta)]_{mn} A^n = e^{if(\theta)} V_\theta A^m V_\theta^{-1} \quad (m = 1, \dots, d).$$

For an injective tensor, clause (2) with *some* V_θ and $f(\theta)$ is a theorem given clause (1) (the “fundamental theorem” of MPS symmetries [1, 2]); the differentiability of the chosen family is the one extra smoothness assumption, labelled (S);

3. (*opposite densities*) the two charge densities are antisymmetric:

$$\rho := \omega_\alpha(S_0^z) = -\omega_\beta(S_0^z).$$

Here ρ is a real parameter of the pair; nothing is assumed about its value, and $\rho = 0$ is allowed;

4. (*stationarity*) both vacua are invariant in time under the dynamics of Definition 2.7, and H commutes with rotations.

Example 3.4. The simplest example is the spin- $\frac{1}{2}$ chain with $\alpha =$ all spins up, $\beta =$ all spins down: product states ($\chi = 1$) with $\rho = \frac{1}{2}$. A genuinely correlated example with $\rho = 0$ is the spin-1 chain state of Affleck, Kennedy, Lieb and Tasaki (the *AKLT state*), an explicit injective MPS with $\chi = 2$. The point of the definitions above is that *nothing* below is restricted to product states.

The first theorem was found as a by-product of the campaign and constrains which densities can occur at all. It has the flavor of the Lieb–Schultz–Mattis-type results, which tie densities of symmetric ground states to lattice constraints, but its proof is a short exact computation.

Theorem 3.5 (density quantization; claim M-IDX-density). *Under Definition 3.3 (clauses 1–3, with the smoothness (S) at both vacua) and Hypothesis 2.5 with constant c :*

$$e^{2\pi i \rho} = c = e^{-2\pi i \rho}, \quad \text{and therefore} \quad 2\rho \in \mathbb{Z}.$$

Proof sketch. Differentiating the local-implementation identity of Definition 3.3(2) at $\theta = 0$ and taking the expectation value in ω_α gives $f'(0) = \omega_\alpha(S_0^z) = \rho$: the phase slope *is* the density. The identity composes under successive rotations, and injectivity makes the decomposition into (f, V_θ) rigid enough that f is additive; with differentiability this forces $f(\theta) = \rho\theta$ up to integer multiples of 2π . At $\theta = 2\pi$ the rotation $u(2\pi) = c\mathbf{1}$ is a pure number by Hypothesis 2.5, so $e^{2\pi i \rho} = c$. Running the same argument in the second vacuum, whose density is $-\rho$, gives $e^{-2\pi i \rho} = c$. Dividing the two conclusions yields $e^{4\pi i \rho} = 1$. The antisymmetry of clause (3) is load-bearing: one vacuum alone gives only $\rho \in \kappa + \mathbb{Z}$, and a general pair only $\rho_\alpha - \rho_\beta \in \mathbb{Z}$. No step uses any relation between ρ and the on-site dimension d . The full proof, in machine-auditable numbered steps, is in the project file `theory/memory-index.md`; its two load-bearing computations carry machine-checked certificates (Section 9). $\square$

4 The wall, the window charge, and the wall position

Definition 4.1 (kink states). A *kink state* (or wall state) is a state ω of the chain that agrees with ω_α far to the left and with ω_β far to the right, with summable charge deviations:

$$\sum_{x < 0} |\omega(S_x^z) - \rho| + \sum_{x > 0} |\omega(S_x^z) + \rho| < \infty.$$

When first moments are needed (they are, in Hypothesis 6.3(iii)) we additionally require the sums above to converge with an extra weight $|x|$, i.e. $\sum_x |x| |\omega(S_x^z) \mp \rho| < \infty$ with the sign matching the corresponding side. From here on, fix a kink state, realize it on a Hilbert space as in Definition 2.3, and call the realizing vector Ψ .

For $\rho > 0$ the charge profile of a kink state falls from $+\rho$ on the far left to $-\rho$ on the far right; the crossover region is the *wall*. Moving the entire wall one site to the right converts one site's expected charge from $-\rho$ to $+\rho$: the region of density $+\rho$ grows by one site, so the total charge *increases* by 2ρ . This calibration — one site of wall motion equals 2ρ units of charge — is the entire mechanism of the memory effect, and motivates:

Definition 4.2 (window charge and wall position). For a finite interval $W = [a, b]$ and a marked interior site $c \in W$ (the *cut*), the *window charge* is the observable

$$\hat{Q}_{W,c} := \sum_{x=a}^c (S_x^z - \rho) + \sum_{x=c+1}^b (S_x^z + \rho),$$

i.e. the charge of W measured against the profile of a reference wall sitting exactly at the cut. The associated *wall-position observable* is

$$X_{W,c} := c + \frac{\hat{Q}_{W,c}}{2\rho} \quad (\rho > 0),$$

so that a state whose wall sits n sites to the right of c (and which is exactly vacuum elsewhere in W) has $\langle X_{W,c} \rangle = c + n$.

Remark 4.3. $\hat{Q}_{W,c}$ is a bounded observable supported in the finite window W ; no infinite sums are involved. All limits ($t \rightarrow \pm\infty$ first, then $W \uparrow \mathbb{Z}$) are taken later, in a stated order, and the order matters.

5 What “memory” means, and why the naive definition fails

The physical effect: prepare a kink plus an incoming localized spin-wave packet (a *magnon*: a traveling disturbance of the vacuum carrying one unit of charge); let them collide; after the collision the wall sits, on average, somewhere else. The displacement persists after the magnon has left — the chain remembers. The question is what exactly is quantized.

One is tempted to define a displacement *operator* “ $X(+\infty) - X(-\infty)$ ” and ask for its spectrum. Two theorems below say this temptation must be resisted, for two independent reasons.

First reason: differences of lattice-valued observables are not lattice-valued. The window charge at late times, $\alpha_t(\hat{Q}_{W,c})$, and at early times, $\alpha_{-t}(\hat{Q}_{W,c})$, do not commute with one another. For noncommuting operators, even when each has spectrum in $\mathbb{Z}$, the difference need not: take, on $\mathbb{C}^2$ with orthonormal basis $\{|0\rangle, |1\rangle\}$ and $|+\rangle := (|0\rangle + |1\rangle)/\sqrt{2}$, the projections $Q_- = |0\rangle\langle 0|$ and $Q_+ = |+\rangle\langle +|$. Both have spectrum $\{0, 1\} \subset \mathbb{Z}$, yet

$$\text{spec}(Q_+ - Q_-) = \left\{ -\frac{1}{\sqrt{2}}, +\frac{1}{\sqrt{2}} \right\},$$

which contains no integers at all. Any correct quantization statement must therefore say *how* the two charges are measured, not merely subtract them.

Second reason: the global charge operator does not exist. One might instead try to build a single conserved operator (“total charge relative to the wall”) and use its spectrum. This fails:

Theorem 5.1 (no global charge operator; claim M-INDEX-LA-strong, refuted form). *There is no construction of a self-adjoint operator $\hat{Q}$ on the Hilbert space of Definition 4.1 that (i) is the limit of the window charges $\hat{Q}_{W,c}$ as $W \uparrow \mathbb{Z}$ in any of the standard operator senses, and (ii) exists for every kink state satisfying the summability condition of Definition 4.1. Two independent obstructions were established:*

1. (fluctuations) *the summability condition controls the means $\omega(S_x^z) \mp \rho$ but not the fluctuations; there are explicit product states (bond dimension 1, zero Hamiltonian) satisfying Definition 4.1 for which the variance of $\hat{Q}_{W,c}$ diverges logarithmically as W grows, so the sequence converges to no operator;*
2. (correlated vacua) *whenever the bond matrices V_θ of Definition 3.3(2) are not multiples of the identity — the generic situation for $\chi \geq 2$ — the half-infinite partial sums of $(S_x^z - \rho)$ fail to converge on the kink vector, by the same mechanism that makes a rotation of half the chain impossible to implement by a norm-convergent sequence of local unitaries.*

Consequently any quantization theorem must be a statement about measurement outcomes, not about the spectrum of a global observable. A weaker existence statement — a charge operator built from the symmetry group acting on the specific Hilbert space realization of the kink state — remains plausible and is recorded as a conjecture in Section 10.

6 The two-measurement protocol and the relaxation hypothesis

The correct object is the outcome statistics of an explicit, physically performable protocol.

Definition 6.1 (projective measurement of the window charge). By Theorem 7.1 below, $\hat{Q}_{W,c}$ has a discrete set of eigenvalues. *Measuring $\hat{Q}_{W,c}$ at time t* means the textbook projective measurement of the Heisenberg observable $\alpha_t(\hat{Q}_{W,c})$: the outcome is an eigenvalue q , obtained with probability $\|E_{W,t}(\{q\})\Psi\|^2$, where $E_{W,t}$ denotes the projection onto the eigenspace of $\alpha_t(\hat{Q}_{W,c})$ with eigenvalue q ; after the measurement the state is the normalized projection.

Definition 6.2 (the two-measurement protocol and its outcome law). Fix the window W , the cut c , an early time $-t_-$ and a late time t_+ (both large). The protocol: (i) measure $\hat{Q}_{W,c}$ at time $-t_-$, record the outcome q_- ; (ii) let the system evolve; (iii) measure $\hat{Q}_{W,c}$ at time t_+ , record q_+ ; (iv) record the *escaped charge*

$$\nu := q_- - q_+,$$

the amount of charge that left the window between the two measurements. The *outcome law* $p_{W;t_-,t_+}(\nu)$ is the probability of recording the value ν , computed by the standard rules of sequential projective measurement:

$$p_{W;t_-,t_+}(\nu) = \sum_q \|E_{W,t_+}(\{q - \nu\}) E_{W,-t_-}(\{q\}) \Psi\|^2.$$

Note what the protocol does *not* assume: nothing about reflected or transmitted waves, nothing about how many kinds of excitation exist, nothing about bound states. It is a pair of measurements and bookkeeping.

The single dynamical assumption of the main theorem says that these statistics settle down at late times and do not leak to infinity as the window grows. “Cesàro convergence” below means convergence of running time-averages: a function $g(t)$ Cesàro-converges if $\frac{1}{T} \int_T^{2T} g(t) dt$ converges as $T \rightarrow \infty$; this is weaker than convergence of $g(t)$ itself and tolerates persistent oscillations.

Hypothesis 6.3 (local relaxation, (LR)). For the fixed kink-plus-packet state Ψ , the fixed cut c , and a fixed growing family of windows $W_m \uparrow \mathbb{Z}$ each containing c :

- (i) (*settling*) there is one sequence of times $T_n \rightarrow \infty$ along which, for every fixed window W , both the single-time expectation values of all observables in W (at times $\pm t$, Cesàro-averaged) and the two-measurement outcome laws $p_{W;t_-,t_+}$ (Cesàro-averaged over both times) converge; call the limiting law p_W ;
- (ii) (*vanishing back-action on the mean*) the first measurement does not shift the late-time mean charge in the limit: the Cesàro average of $\langle \mathcal{D}_{W,-t_-}(\alpha_{t_+}(\hat{Q}_{W,c})) - \alpha_{t_+}(\hat{Q}_{W,c}) \rangle_\Psi$ tends to zero for every fixed W , where $\mathcal{D}_{W,-t_-}(A) := \sum_q E_{W,-t_-}(\{q\}) A E_{W,-t_-}(\{q\})$ is the operation of performing the first measurement and forgetting its outcome. Only this averaged, first-moment statement is assumed — not that the operators commute;

- (iii) (*no charge stranded at the window edge*) the limiting laws are tight with first moments, uniformly in the window:

$$\lim_{M \rightarrow \infty} \sup_m \sum_{|\nu| > M} (1 + |\nu|) p_{W_m}(\nu) = 0.$$

Optionally, one may also assume the laws p_{W_m} converge (weakly, i.e. probability by probability) to a law p ; this optional clause buys only the uniqueness of the limit values and nothing in the quantization statements.

Remark 6.4 (status of (LR)). Hypothesis 6.3 is *assumed*, not derived from the microscopic dynamics; this is stated prominently in Section 10. Its role is to be the *weakest* sufficient condition found under which the quantization argument closes. It is strictly weaker than any scattering assumption: it does not name channels, does not require wave operators, and tolerates bound states, absorption, and any number of excitation species. It can fail — e.g. if charge accumulates permanently at the window boundary — and the numerical probe of Section 9 was designed to look for exactly such failures (none was found).

7 The quantization theorems

Theorem 7.1 (finite-window lattice rigidity; claim M-INDEX-fin). *Assume Hypothesis 2.5 and Definition 3.3(3). Fix any finite window $W = [a, b]$ and cut $c \in W$. Then:*

1. *all eigenvalues of $\hat{Q}_{W,c}$ lie in a single shifted integer lattice $\kappa_{W,c} + \mathbb{Z}$, where the shift $\kappa_{W,c} \equiv |W|\kappa + \rho(a + b - 1 - 2c) \pmod{\mathbb{Z}}$ depends on the window, the cut, and the density, but not on the state and not on time (time evolution preserves the spectrum of an observable);*
2. *consequently, in the protocol of Definition 6.2, both recorded outcomes $q_{\pm}$ lie in the same shifted lattice, the shifts cancel in the difference, and every recorded escaped charge $\nu = q_- - q_+$ is an exact integer — at every window size, at all times, with no hypotheses on the dynamics beyond its existence.*

Proof sketch. Each S_x^z has spectrum in $\kappa + \mathbb{Z}$ (Hypothesis 2.5); the window charge is a sum of commuting such operators minus the number $\rho(c - a + 1) - \rho(b - c)$; adding commuting operators adds their spectra. The integrality of ν is then arithmetic — crucially, arithmetic of two outcomes of the *same* observable at fixed W , so the state-independent shift cancels exactly. This is where the protocol earns its keep: it is immune to the noncommutativity trap of Section 5 because no operator subtraction ever occurs, only subtraction of recorded numbers. $\square$

Theorem 7.2 (quantization of the memory outcome law; claim M-INDEX-spec). *Assume the setting of Sections 2–4 (finite-range charge-conserving dynamics, symmetric vacuum pair with density $\rho > 0$, kink state with summable deviations and first moments), Hypothesis 2.5, and Hypothesis 6.3. Then, along the time sequence provided by (LR):*

1. *for every fixed window W_m the limiting outcome law p_{W_m} exists and is supported on the integers;*
2. *every limit point $p = \{p_{\nu}\}_{\nu \in \mathbb{Z}}$ of the family $\{p_{W_m}\}$ as $m \rightarrow \infty$ is a probability distribution supported on $\mathbb{Z}$: no probability escapes to infinity and none lands off the integer lattice. (With the optional convergence clause of (LR) the limit p is unique.);*

3. the asymptotic mean wall displacement, defined as the ordered double limit (time limits first at fixed window, then $W_m \uparrow \mathbb{Z}$) of the difference of expectations of the wall-position observable $X_{W,c}$ of Definition 4.2, exists along the same subsequence and equals

$$\delta x = -\frac{1}{2\rho} \sum_{\nu \in \mathbb{Z}} \nu p_\nu$$

4. in particular, every individual outcome of the protocol displaces the recorded wall position by an exact integer multiple of $1/(2\rho)$ lattice sites, while the mean δx is a probability-weighted average of these quantized outcomes and is not itself quantized.

Bound states, absorption of the magnon by the wall, additional excitation species, and any other exotic process are all permitted: they contribute probability at other integers ν and cannot produce non-integer outcomes.

Proof sketch. Clause (1): Theorem 7.1(2) puts each finite-time law on $\mathbb{Z}$; Cesàro limits of probability laws on the closed set $\mathbb{Z}$ stay on $\mathbb{Z}$ provided no mass escapes, which at fixed W is automatic (finitely many eigenvalues). Clause (2): as W grows the lattice itself is fixed (integers), and Hypothesis 6.3(iii) is exactly the tightness needed so that weak limit points remain probability laws on $\mathbb{Z}$ (this is the standard compactness argument for tight families, and the counterexamples showing that plain convergence of means would *not* suffice are recorded alongside the proof). Clause (3): at fixed window and finite times, charge conservation (Definition 2.7) gives an exact identity: the change of $\langle \hat{Q}_{W,c} \rangle$ equals the net current through the two ends, and the first-moment identity $\sum_\nu \nu p_{W;t_-,t_+}(\nu) = \langle \alpha_{-t_-}(\hat{Q}) \rangle - \langle \mathcal{D}_{W,-t_-}(\alpha_{t_+}(\hat{Q})) \rangle$ relates the protocol mean to the unmeasured evolution up to the back-action term that Hypothesis 6.3(ii) kills in the limit; Hypothesis 6.3(iii) (first moments, not just tightness) lets the mean pass through the window limit. Dividing by -2ρ converts escaped charge to wall displacement (Definition 4.2). The full numbered-step proof is `theory/memory-index.md`; the first-moment identity and the tightness counterexamples carry machine certificates. $\square$

Remark 7.3 (consistency with the scattering picture). Suppose one *additionally* assumes the full scattering package that earlier work had to assume from the start: a complete two-channel scattering theory in which the incoming magnon (charge -1 relative to the left vacuum) either reflects (outgoing charge -1 on the same side, escaped charge $\nu = 0$) or transmits (outgoing charge $+1$ relative to the right vacuum, escaped charge $\nu = 2$), with transmission probability T . Then the outcome law of Theorem 7.2 collapses to $p_0 = 1 - T$, $p_2 = T$, and the boxed formula gives $\delta x = -T/\rho$: exactly the previously known conditional result. The logical relation is strict: the old theorem = the new theorem + the scattering package; the quantization itself never needed the package.

8 Existence of scattering states from the excitation ansatz

Theorem 7.2 quantizes the outcomes without knowing their probabilities. Computing the probabilities p_ν is the job of scattering theory, and the obstruction there is well known: proving that wave operators exist and are complete for a many-body lattice system is out of reach in general, and completeness can even be false as posed (extra channels, bound states). The final theorem shows that the *existence* half — enough to define the transmission probability on the

states the construction reaches — follows from hypotheses that the matrix-product *excitation ansatz* [3, 4, 5] is designed to provide. The philosophy: the ansatz supplies the kinematic data (which single excitations exist and how they disperse); once those data are assumed exact, the dynamical statement (scattering states exist) becomes a theorem. Completeness is *not* claimed, and by Theorem 7.2 it is no longer needed for quantization.

Definition 8.1 (wave operators). Let $\mathcal{H}$ carry the dynamics e^{-iHt} of the kink sector (the Hilbert-space realization of Definition 2.3 for kink states), and let $\mathcal{H}_{\text{ch}}$ be a *channel space*: square-summable functions of a kink momentum p and a magnon momentum k , evolving under multiplication by $e^{-i(E_K(p)+\omega(k))t}$, where E_K and ω are given functions (the *dispersions*: the energy of each excitation as a function of its momentum; the derivative of a dispersion is the excitation’s propagation speed, its *group velocity*). Given a linear *identification map* $J : \mathcal{H}_{\text{ch}} \rightarrow \mathcal{H}$ (built below from the ansatz data), the *wave operators* are the strong limits

$$W_{\pm} := \text{s-lim}_{t \rightarrow \pm\infty} e^{iHt} J e^{-iH_{\text{ch}}t}$$

whenever they exist on a stated domain. Physically: $W_- \phi$ is the true state of the interacting chain that, in the far past, looked like the freely moving kink-plus-magnon configuration ϕ .

Hypothesis 8.2 (exact band data; abbreviated statement). There exist: (a) a family of *kink vectors* $\kappa_a(x) \in \mathcal{H}$ ($x \in \mathbb{Z}$, finitely many labels a), translation-covariant, whose closed span is invariant under H with matrix dispersion E_K , twice continuously differentiable in momentum; (b) over each vacuum, *magnon creation operators*: almost-local operators (localized up to tails decaying faster than any power) adding one unit of charge, whose action on the vacuum realizes an isolated single-magnon energy band with dispersion ω and positive gap; both band relations hold *exactly*, not approximately — this is the mathematical idealization of “the excitation ansatz is exact”; (c) a *two-cluster factorization estimate*: there are $C < \infty$ and $\lambda \in (0, 1)$ such that for all kink vectors and all observables A localized at least r sites to the left and B at least r sites to the right of the kink positions,

$$|\langle \kappa_a(x), AB \kappa_{a'}(x') \rangle - \omega_a(A) \omega_b(B) \langle \kappa_a(x), \kappa_{a'}(x') \rangle| \leq C \|A\| \|B\| \lambda^r;$$

(d) the incoming wave packets are smooth with disjoint group-velocity supports separated by some $\varepsilon_v > 0$ (in particular, this is a fixed-packet theorem: the zero-velocity limit $k \rightarrow 0$ is excluded, and no uniformity in it is claimed).

Theorem 8.3 (existence of kink–magnon scattering states; claim AC-EX, parts ACE.1–ACE.2). *Under Definition 3.3, Definition 2.7 and Hypothesis 8.2, with the identification map J built by applying a momentum-filtered magnon creation operator to a kink vector, the wave operators W_- (one incoming configuration) and W_+ (two outgoing configurations, magnon on the left or on the right of the kink) exist on the stated packet domains and are isometries (norm-preserving maps, so no probability is lost in the construction). On the constructed outgoing subspace, the orthogonal projection N_T^{ex} onto the transmitted configuration is well defined, and $\langle \Psi, N_T^{\text{ex}} \Psi \rangle$ is the transmission probability of the constructed channels. Nothing is asserted about the orthogonal complement of the constructed subspace: completeness — the statement that the constructed states exhaust everything that propagates — is not claimed, is not used anywhere in this report, and is known to be unprovable at this generality.*

Proof sketch. The standard route (Cook’s method): show that the time derivative of $e^{iHt} J e^{-iH_{\text{ch}}t} \phi$ is integrable in t . The derivative is controlled by (i) the exactness of the band relations, which

makes the naive error term vanish identically for a single excitation, leaving only cross terms between the kink and the magnon; (ii) the cluster estimate (c) and the velocity separation (d), which confine the cross terms to an exponentially suppressed spatial cone, giving an integrable $O(|t|^{-3})$ bound. A technical device is essential and easy to state: the magnon creation operators must be *momentum-filtered* (their momentum content cut off smoothly around the packet support) — with the naive unfiltered normalization the hopping kernels decay too slowly and the integral does not close. The port of this argument from the translation-invariant setting of [6] to the kink sector (where the wall breaks translation invariance of the *state* but the sector as a whole still carries a translation action) is the substance of the proof; every step is re-derived rather than cited by analogy. Full proof: `theory/ansatz-scattering.md`. $\square$

Remark 8.4 (bridge to the outcome law; conditional). On states in the range of both wave operators, one expects the outcome law of Theorem 7.2 to be carried by the constructed channels with $p_2 = \langle N_T^{\text{ex}} \rangle$ (as in Remark 7.3). This bridge is proved *conditionally* on three explicitly stated extra assumptions: the charge bookkeeping of the two legs (q -values $-1, -1, +1$ as in Remark 7.3), a local-decay hypothesis (the outgoing state, restricted to any fixed window around the wall, becomes charge-definite — the missing lemma is named and tracked), and the applicability of Hypothesis 6.3 to the vector in question. None of the three is proved here; see Section 10.

9 Numerical verification

Three independent machine checks accompany the theorems. Two conventions first. *Exact diagonalization* (ED) means: represent the full many-body Hilbert space of a finite chain on a computer with no approximation beyond machine arithmetic, and evolve by the exact matrix exponential. A *red test* (used throughout the project) means: every machine-checked certificate must be demonstrated to *fail* when one of its hypotheses is deliberately broken; a check that cannot fail certifies nothing. All checkers pass their green runs and fail their red runs; the suite is rerun before every status change.

Pre-registered falsification attempt. Before the proofs of Section 7 existed, a falsifier was specified: if the outcome distribution of the window charge, in an honest simulation of a kink–magnon collision, put persistent probability off the integer lattice or leaked mass under growing windows, the theorem targets would have been withdrawn. The model: the spin- $\frac{1}{2}$ easy-axis XXZ ferromagnet,

$$H = -J \sum_x \left[S_x^x S_{x+1}^x + S_x^y S_{x+1}^y + \Delta (S_x^z S_{x+1}^z - \frac{1}{4}) \right], \quad \Delta = 2, J = 1,$$

with boundary spins frozen up (left) and down (right) to hold one wall in the system; here $\rho = \frac{1}{2}$, so the predicted displacement quantum is $1/(2\rho) = 1$ site per unit of escaped charge, and the transmission outcome ($\nu = 2$) displaces by 2 sites. A chain of $N = 50$ sites (restricted to at most five domain walls, a truncation validated against the exact space at $N = 22$ to 3.4×10^{-3} in every probability) was prepared with a *dressed* kink (the true ground state of the kink sector, not a sharp product wall) and an incoming magnon packet, and evolved through the collision. Results: the probability off the integer lattice was *exactly zero* to machine precision at every time, window, and geometry tested; at the latest clean time the law concentrated on the two scattering outcomes, $p(\nu = 2) = 0.9996$, $p(\nu = 0) = 0.0004$ (this model is nearly transparent at the chosen momentum), with residual mass $\leq 2.5 \times 10^{-3}$ on neighboring integers; the mean displacement computed from

the outcome law agreed with an independent estimator of the wall position (a profile centroid, algebraically unrelated to the window charge) to 0.011 sites. Two instructive artifacts were found and documented: preparing a *sharp* wall instead of the dressed one fakes a 5% off-outcome signal (the preparation defect radiates slow magnon pairs — a state-preparation effect, not a violation); and the residual integer-valued mass sits in near-threshold two-magnon configurations with group velocity near zero — precisely the slow-excitation regime that scattering-based approaches cannot control, absorbed here by Theorem 7.2 because such outcomes still land on integers.

Certificates for the proofs. A second program certifies, in exact or high-precision arithmetic, the load-bearing computations of Sections 3–7: the lattice arithmetic of Theorem 7.1 including an irrational-offset stress test; the 2×2 counterexample of Section 5; the divergent-fluctuation counterexample of Theorem 5.1; the density-equals-phase-slope computation behind Theorem 3.5 on randomly drawn symmetric tensors of bond dimensions up to 4 (agreement to 9×10^{-16} , with the necessity of the antisymmetric vacuum pair red-certified); the first-moment identity of Theorem 7.2; and the exact reduction of Remark 7.3. A third program certifies the two quantitative pillars of Theorem 8.3: the band and threshold data of the reference model, and the decay rate of the Cook integrand, with a red mode confirming that a slowly decaying (unfiltered) kernel destroys the required integrability — the failure mode the momentum filter exists to prevent.

10 What is not proved

For clarity, the complete list of assumptions and open items.

1. *Hypothesis (LR) is not derived from the dynamics.* No proof is offered that any concrete Hamiltonian satisfies the relaxation clauses; (LR) is the stated interface between the quantization arithmetic and the hard open problems of many-body relaxation theory. In particular, no claim of the form “scattering assumptions imply (LR)” is made; an attempted proof of such an implication was reviewed and rejected during verification.
2. *Hypothesis 8.2 (exact band data) is not derived.* In practice the excitation ansatz provides these data as the output of a numerical optimization; here their exactness is an assumption. The cluster estimate (c) is the load-bearing clause and is verified on no model.
3. *Completeness of scattering is neither proved nor assumed.* It is provably not derivable from band data alone, and Theorem 7.2 was engineered precisely so that it is not needed.
4. *The bridge of Remark 8.4 rests on a named, unproved local-decay lemma and on (LR); both are tracked as open problems.*
5. *The mean δx is not quantized,* and no claim to the contrary is made anywhere; quantization is a property of individual protocol outcomes.
6. *Conjecture (charge operator on the kink representation).* While Theorem 5.1 kills the state-independent construction, it is conjectured that on the specific Hilbert-space realization of a fixed kink state the symmetry rotations are implemented by a continuous unitary group whose generator has purely discrete, integer-spaced spectrum; supporting evidence (uniformly bounded charge variance on such realizations) is recorded, but no proof exists.

7. *Model scope of the numerics.* The simulations concern one ferromagnetic model family; they are consistency checks and falsification attempts, not proofs, and are labelled as such throughout.

A How these results were verified

Each theorem above was produced by an adversarial loop: a *proposer* (an automated prover) writes the full argument in numbered, individually addressable steps, every step justified by a numbered definition, a previously established claim, or a named machine-checked computation; an independent *critic* (a different automated system, from a different model family, to decorrelate blind spots) attacks the argument by recomputing its key steps, constructing counterexamples, and auditing every quantifier, and issues a written verdict listing objections classified by severity, each with a demanded fix and a statement of what survives if the fix fails. The loop repeats until a verdict contains no fatal or major objection. For the results reported here the loop ran three rounds; the objection counts fell from 20 (first round, three parallel verdicts) to 14 to 13, with major objections $6 \rightarrow 3 \rightarrow 1 \rightarrow 0$ and no fatal objection at any round. Two of the corrections flowed in the reverse direction — errors in a critic’s verdict caught and refuted by the machine-checked certificates — which is the intended behavior of the design. Two draft claims did not survive: the global charge operator (now the refutation, Theorem 5.1) and an implication from scattering assumptions to (LR) (withdrawn; item 1 of Section 10). The verdicts, proofs, and certificates are in the project repository (`theory/memory-index.md`, `theory/ansatz-scattering.md`, `theory/verdicts/memory-index-r1-r3`, `theory/checks/`).

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